Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Paper 6: Memory Production Review Response

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Mon, Sep 21, 2026 at 10:55 PM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Ben,

Thank you for your detailed response to Sara's and my independent reports.
I have compared your response with the new version of the manuscript and re-derived the principal new arguments independently.
The revision addresses the substantive discrepancies we identified between the statements of the results and their proved scope.
My assessment of this paper, following this response cycle, is favourable, subject to the qualifications below.

Points resolved in the revised manuscript

Section 2.1 now states explicitly that the metric (6) lies outside the companion's source-weighted admissible class, and supplies the equivalence constants; these are immediate, since the ratio of weights is 1/p_j(v).
Lemma 2.5 and Proposition 2.6 give a direct comparison between the dimension-matched charge and the deficit norm in the stated law metric,
which removes the dependence at d = n altogether.
The abstract, Corollary 4.8 and the conclusion now distinguish the exact deficit identities from the resulting bounds on the charge,
and the lower bound of Corollary 4.8, restated as ‖Φ + P‖/L*, is no longer vacuous in the mixed regime.

The revision separates the two-time Chapman–Kolmogorov condition from Markovianity of the observed process.
Corollary 4.6 is appropriately renamed, and Example 4.7 supplies a valid flat fine chain whose observed bits are pairwise independent
and whose three-time law is not Markov. The same example distinguishes full dynamical closure from closure on the accessible subspace,
and Remark 6.8 makes the same limitation explicit for the interference construction. You were right that the corollary previously had no
witness for that non-implication under its own hypothesis; neither report had noticed.

Theorem 5.2 states its flatness hypothesis and the restricted class of block-supported stochastic sections, and the manuscript no longer presents
this as uniqueness among arbitrary Mori or Wiener projections.
Remark 5.3 limits canonicality to h_n, and the proof of Example 5.4 establishes the minimum and the two minimisers for the family it constructs.

The remaining changes also respond to the reports: production is assigned no sign; the status of the law-weighted geometry and of the
Mori–Zwanzig analogy is explicit; the tower result is called monotonicity of the deficit norm; and Table 2 supplies exact charge intervals.
Your decision not to report numerical minimisers is acceptable, because the manuscript now states what the intervals establish and does
not present them as computed charges (and Example 5.4 in fact supplies one charge exactly, $h_2 = \sqrt{2}$).

Verification of the new material

Since none of this was before either report, I re-derived it rather than accepting the replication record.
Lemma 2.5 held on $2 \cdot 10^5$ random triples, with worst ratios 0.938 in (a) and 0.9992 in (b), so (b) is essentially sharp.
The constants of Proposition 2.6 were not violated in trials on random families with m = 3, 4, 5 and N = 2, 3, 4.
All three windows of Table 2 recompute exactly: 54/456125 with $L^{*2} = 445/69$, 459/6732200 with $L^{*2} = 821/69$, and 9/28672 for Proposition 6.5.
My four-parameter minimisation for Example 3.2 gives $h_2 = 0.0265026$, inside the window.
Example 5.4's central identity is exact and its equality conditions give precisely your two families.
Example 4.7 checks in every particular, including $\|LV(2 \leftarrow 1)Q_1\| = 1/2$ on the full space against vanishing on A_1.
The dial intervals recompute from BA and $\iota_0[\delta_0]$ to [−5/32, 5/16] and [−3/32, 3/16].

Qualifications and requests for the record

1. Example 5.4 uses a family that the manuscript correctly identifies as not process-induced. It therefore establishes non-uniqueness of nearest flat families
   in the full space of families carrying laws, not within the process-induced class. Please retain that distinction wherever the example is cited.
   I would also ask you to pose the restricted question explicitly: can the nearest flat family fail to be unique for a *process-induced* family?
   The programme's objects are process-induced, and a negative answer would recover the canonical representative exactly where it is wanted.
   This is a scope qualification and does not affect the balance law.
2. The lower bound of Proposition 2.6 appears to be weaker than your own argument delivers, in the case that matters.
   The middle term is bounded by Lemma 2.5(b) with K = V(r ← s); for a process-induced family that kernel transports the laws,
   so its distortion is exactly 1 and the factor 1/min p_r is not needed. On the three-point grid this would give $L^{*2} = 2 + 1/\min_v p_2(v)$ rather than
   $1 + 1/\min_w p_1(w) + 1/\min_v p_2(v)$, improving Example 3.2 from 445/69 to 110/23 and the lower bound from 0.01088 to $\sqrt{(9/56375)} = 0.01264$.
   Random process-induced families are consistent with the sharper constant. The general statement plainly needs the printed one, since a family may
   carry laws without transporting them (Example 5.4 being your own case in point) so this would be a corollary for the process-induced case rather than a correction. It matters because L* is what makes Table 2 uninformative: the windows span factors of 2.5 to 3.4, and in the one case where the true value
   is known it sits at 96% of the upper end. On the other side, C* looks slack for m ≥ 3, since the upper-bound argument uses only the consecutive-cut
   triples while bounding against the full deficit cochain; in trials on four- and five-point grids I never saw h_n/‖Δ_V‖ exceed 0.976, with no visible
   degradation in ρ. Can C* = 1 be proved on every grid?
3. Remark 2.4's claim that Lemma 2.3 fails for the source-weighted class is the only structural assertion in the paper without an exact counterexample,
   which is the standard §7 sets for everything else. One explicit rational family failing by an exact amount would close it in the paper's own register.
   I would not print a percentage range: in my own sampling the relative residual has median 7.9% and maximum 63.5%, and those figures depend
   on the sampling law, as the difference between my numbers, yours and the ones in my report shows.
4. Remark 2.7 and open problem (ii) explain why the argument stops at d = n. It is worth stating what follows: at d = n the charge and ‖Δ_V‖ are
   bi-Lipschitz equivalent, so the distinctive content of the charge as an invariant resides in the d > n profile.
5. Minor: the Lebesgue-null claim in Remark 5.3 is attributed to the companion's first-order theory and to variational analysis, but the fact is
   classical and stronger (for any closed set in Euclidean space the points with a non-unique metric projection form a null set) and needs neither.

One check outstanding on my side

Remark 2.7 states that Proposition 2.6 is the comparison [21, Proposition 5.3] at d = n with exponent one.
Since [21] is itself in the submitted portfolio and carries outstanding observations from its own technical evaluation,
I will verify revision 2 of that paper and the consistency of the two statements before integrating.
Nothing in that check bears on the assessment above.

I regard the questions raised in my technical report as answered for the purposes of this paper's examination.

Best,
Ezequiel

[Quoted text hidden]